

1 Gain law

(bars denote normalized quantities: lengths by R ; $a_o := \Delta t / (\gamma_N R)$ [1/pN] and $a_{\text{nom}} := \Delta t / \gamma_N = a_o R$ [$\mu\text{m}/\text{pN}$]; $\bar{a} := a / a_{\text{nom}}$; forces by a_o : $\bar{f} := a_o f$, so that $\bar{a} \bar{f} = \Delta \bar{w}$; $0 \leq \bar{a} \leq 1$, far field 1, contact 0; hats sit on the normalized quantities $\hat{a}_w, \hat{b}, \delta \hat{w}$; $e_\theta = \theta - \hat{\theta}$)

$$\begin{aligned}\bar{a}(\bar{w}) &= 1 - \frac{1}{b(\bar{w} - \bar{w}_s)} \\ \bar{a}'(\bar{a}, b) &:= \frac{d\bar{a}}{d\bar{w}} = b(1 - \bar{a})^2 \\ \bar{a}''(\bar{a}, b) &:= \frac{d\bar{a}'}{d\bar{w}} = -2b^2(1 - \bar{a})^3 \\ a_w &= a_o \bar{a}_w, \quad a_o = \frac{\Delta t}{\gamma_N R}\end{aligned}$$

2 True dynamics

$$(\delta \bar{w}_j[k] := \delta \bar{w}[k - d + j - 1], \quad j = 1, \dots, d + 1, \quad d = 2)$$

$$\begin{aligned}\bar{w}[k] &= \bar{w}_d[k] - \delta \bar{w}_3[k] \\ \bar{w}_d[k + 1] &= \bar{w}_d[k] + \Delta \bar{w}_d[k] \\ \delta \bar{w}_1[k + 1] &= \delta \bar{w}_2[k] \\ \delta \bar{w}_2[k + 1] &= \delta \bar{w}_3[k] \\ \delta \bar{w}_3[k + 1] &= \lambda_c \delta \bar{w}_3[k] - F_{dw}[k] e_{\bar{a}_w}[k] - \varepsilon_w[k] \\ \Delta \bar{w}[k] &= \Delta \bar{w}_d[k] + (1 - \lambda_c) \delta \bar{w}_3[k] + F_{dw}[k] e_{\bar{a}_w}[k] + \varepsilon_w[k] \\ \bar{a}_w[k + 1] &= \bar{a}_w[k] + b[k] (1 - \bar{a}_w[k])^2 \Delta \bar{w}[k] \\ b[k + 1] &= b[k]\end{aligned}$$

$$\begin{aligned}F_{dw}[k] &= \bar{f}_{dw}[k] + (1 - \lambda_c) \sum_{i=1}^d \bar{f}_{dw}[k - i] \\ \varepsilon_w[k] &= \bar{a}_w[k] \bar{f}_T[k] + (1 - \lambda_c) \sum_{i=1}^d \bar{a}_w[k - i] \bar{f}_T[k - i] - (1 - \lambda_c) n_w[k]\end{aligned}$$

3 Estimator

$$\mathbf{x} = [\delta \bar{w}_1, \delta \bar{w}_2, \delta \bar{w}_3, \bar{a}_w, b]^\top \quad (d = 2)$$

$$\begin{aligned}
\delta \hat{w}_1[k+1] &= \delta \hat{w}_2[k] \\
\delta \hat{w}_2[k+1] &= \delta \hat{w}_3[k] \\
\delta \hat{w}_3[k+1] &= \lambda_c \delta \hat{w}_3[k] \\
\Delta \hat{w}[k] &= \Delta \bar{w}_d[k] + (1 - \lambda_c) \delta \hat{w}_3[k] \\
\hat{a}_w[k+1] &= \hat{a}_w[k] + \hat{b}[k] (1 - \hat{a}_w[k])^2 \Delta \hat{w}[k] \\
\hat{b}[k+1] &= \hat{b}[k]
\end{aligned}$$

4 Error dynamics

(True – Estimator; coefficients at the estimate)

$$\begin{aligned}
e_{\Delta \bar{w}}[k] &= \Delta \bar{w}[k] - \Delta \hat{w}[k] = (1 - \lambda_c) e_{\delta \bar{w}_3}[k] + F_{dw}[k] e_{\bar{a}_w}[k] + \varepsilon_w[k] \\
b[k] (1 - \bar{a}_w[k])^2 &= \hat{b}[k] (1 - \hat{a}_w[k])^2 - 2 \hat{b}[k] (1 - \hat{a}_w[k]) e_{\bar{a}_w}[k] + (1 - \hat{a}_w[k])^2 e_b[k] + \mathcal{O}(e^2)
\end{aligned}$$

$$\begin{aligned}
e_{\delta \bar{w}_1}[k+1] &= e_{\delta \bar{w}_2}[k] \\
e_{\delta \bar{w}_2}[k+1] &= e_{\delta \bar{w}_3}[k] \\
e_{\delta \bar{w}_3}[k+1] &= \lambda_c e_{\delta \bar{w}_3}[k] - F_{dw}[k] e_{\bar{a}_w}[k] - \varepsilon_w[k] \\
e_{\bar{a}_w}[k+1] &= \left[1 + \hat{b}(1 - \hat{a}_w)^2 F_{dw} - 2 \hat{b}(1 - \hat{a}_w) (\Delta \bar{w}_d + (1 - \lambda_c) \delta \hat{w}_3) \right] e_{\bar{a}_w}[k] \\
&\quad + (1 - \lambda_c) \hat{b}(1 - \hat{a}_w)^2 e_{\delta \bar{w}_3}[k] + (1 - \hat{a}_w)^2 (\Delta \bar{w}_d + (1 - \lambda_c) \delta \hat{w}_3) e_b[k] + \hat{b}(1 - \hat{a}_w)^2 \varepsilon_w[k] \\
e_b[k+1] &= e_b[k]
\end{aligned}$$

$$F_e[k] = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & \lambda_c & -F_{dw} & 0 \\ 0 & 0 & (1 - \lambda_c) \hat{b}(1 - \hat{a}_w)^2 & 1 + \hat{b}(1 - \hat{a}_w)^2 F_{dw} & (1 - \hat{a}_w)^2 \\ & & & -2 \hat{b}(1 - \hat{a}_w) (\Delta \bar{w}_d + (1 - \lambda_c) \delta \hat{w}_3) & \times (\Delta \bar{w}_d + (1 - \lambda_c) \delta \hat{w}_3) \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\mathbf{q}[k] = \left[0, 0, -\varepsilon_w, \hat{b}(1 - \hat{a}_w)^2 \varepsilon_w, 0 \right]^\top$$

5 Measurement

$$(\nabla_d \bar{w}_d[k] := \bar{w}_d[k] - \bar{w}_d[k - d])$$

$$\begin{aligned}
y_1[k] &= \delta \bar{w}_1[k] + n_w[k] \\
y_2[k] &= \bar{a}_w[k - d] + n_a[k], \quad \bar{a}_w[k - d] = \bar{a}_w[k] - b(1 - \bar{a}_w[k])^2 \nabla_d \bar{w}_d[k] \\
\hat{y}_1[k] &= \delta \hat{w}_1[k] \\
\hat{y}_2[k] &= \hat{a}_w[k] - \hat{b}[k](1 - \hat{a}_w[k])^2 \nabla_d \bar{w}_d[k] \\
H_{24}[k] &= \frac{\partial \hat{y}_2[k]}{\partial \hat{a}_w[k]} = 1 + 2 \hat{b}[k](1 - \hat{a}_w[k]) \nabla_d \bar{w}_d[k] \\
H_{25}[k] &= \frac{\partial \hat{y}_2[k]}{\partial \hat{b}[k]} = -(1 - \hat{a}_w[k])^2 \nabla_d \bar{w}_d[k] \\
H[k] &= \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & H_{24}[k] & H_{25}[k] \end{bmatrix} \\
\mathbf{e}_y[k] &= H[k] \mathbf{e}^-[k] + \begin{bmatrix} n_w[k] \\ n_a[k] \end{bmatrix}
\end{aligned}$$

6 Noise

$$\begin{aligned}
Q[k] &= Q_{33}[k] \mathbf{g}[k] \mathbf{g}[k]^\top, \quad \mathbf{g}[k] = \begin{bmatrix} 0, & 0, & -1, & \hat{b}[k](1 - \hat{a}_w[k])^2, & 0 \end{bmatrix}^\top, \quad Q_{55} = 0 \\
Q_{33}[k] &= \kappa_T \left(\hat{a}_w[k] + (1 - \lambda_c)^2 (\hat{a}_w[k - 1] + \hat{a}_w[k - 2]) \right) + (1 - \lambda_c)^2 \sigma_{n_w}^2, \quad \kappa_T = \frac{4k_B T a_o}{R} \\
R[k] &= \text{diag}(R_1, R_2[k]), \quad R_1 = \sigma_{n_w}^2 = \frac{\sigma_n^2}{R^2}, \quad R_2[k] = 2a_{\text{cov}}^2 IF_{\text{var}}(\hat{a}_w[k]) (\hat{a}_w[k] + \xi)^2
\end{aligned}$$

7 Observability

$$\begin{aligned}
\mathbf{e}[k] &= [e_{\delta\bar{w}_1}, e_{\delta\bar{w}_2}, e_{\delta\bar{w}_3}, e_{\bar{a}_w}, e_b]^\top, \mathbf{n}[k] = [n_w[k], n_a[k]]^\top \\
\Phi(k, k_0) &:= F_e[k-1] F_e[k-2] \cdots F_e[k_0], \Phi(k_0, k_0) = I \\
N &\text{ window length} \\
\mathbf{Y} &:= [\mathbf{e}_y[k_0]; \mathbf{e}_y[k_0+1]; \dots; \mathbf{e}_y[k_0+N]], \mathbf{N} := [\mathbf{n}[k_0]; \mathbf{n}[k_0+1]; \dots; \mathbf{n}[k_0+N]] \\
X_{:,j} &\text{ column } j \text{ of } X
\end{aligned}$$

$$\begin{aligned}
F_e[k]_{:,5} &= \left[0, 0, 0, (1 - \hat{a}_w)^2 (\Delta\bar{w}_d + (1 - \lambda_c) \delta\hat{w}_3), 1 \right]^\top \\
H[k]_{:,5} &= \left[0, -(1 - \hat{a}_w)^2 \nabla_d \bar{w}_d \right]^\top
\end{aligned}$$

$$\begin{aligned}
\mathbf{e}[k] &= \Phi(k, k_0) \mathbf{e}[k_0] \\
\mathbf{e}_y[k] &= H[k] \Phi(k, k_0) \mathbf{e}[k_0] + \mathbf{n}[k]
\end{aligned}$$

$$\mathbf{Y} = O(k_0, N) \mathbf{e}[k_0] + \mathbf{N}, \quad O(k_0, N) := \begin{bmatrix} H[k_0] \Phi(k_0, k_0) \\ H[k_0+1] \Phi(k_0+1, k_0) \\ \vdots \\ H[k_0+N] \Phi(k_0+N, k_0) \end{bmatrix}$$

$$\begin{aligned}
R_w &:= \text{diag}(R[k_0], R[k_0+1], \dots, R[k_0+N]), \quad O_w(k_0, N) := R_w^{-1/2} O(k_0, N) \\
\mathbf{Y}_w &:= R_w^{-1/2} \mathbf{Y} = O_w(k_0, N) \mathbf{e}[k_0] + \mathbf{N}_w, \quad \mathbb{E}[\mathbf{N}_w \mathbf{N}_w^\top] = I
\end{aligned}$$

$$\begin{aligned}
O_w(k_0, N) \mathbf{v}_i &= \sigma_i \mathbf{u}_i, \quad i = 1, \dots, 5, \quad \sigma_1 \geq \dots \geq \sigma_5 \geq 0, \quad \|\mathbf{v}_i\| = \|\mathbf{u}_i\| = 1 \\
\sigma_1 &= \max_{\|\mathbf{v}\|=1} \|O_w(k_0, N) \mathbf{v}\|, \quad \sigma_5 = \min_{\|\mathbf{v}\|=1} \|O_w(k_0, N) \mathbf{v}\|
\end{aligned}$$

$$\text{rank } O_w(k_0, N) = \#\{i : \sigma_i > 0\}$$

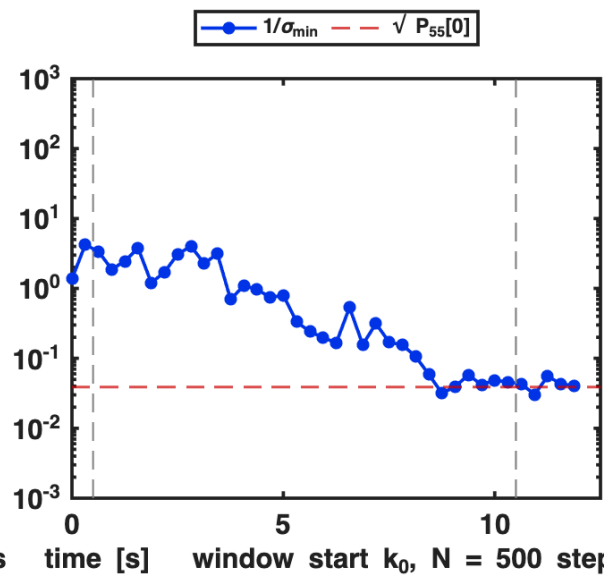
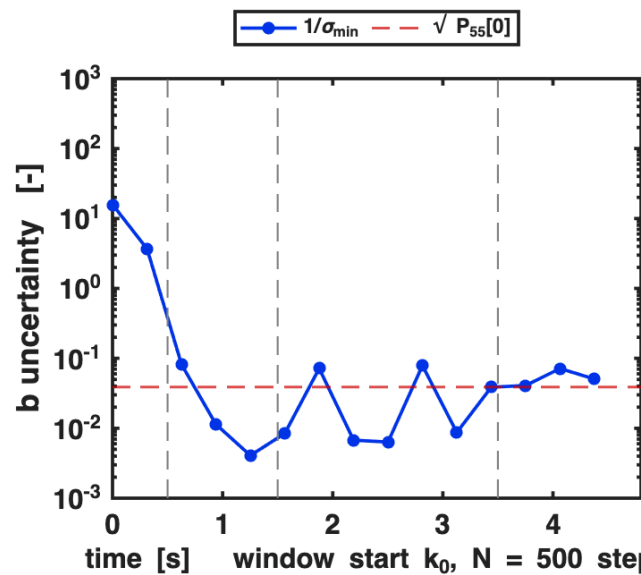
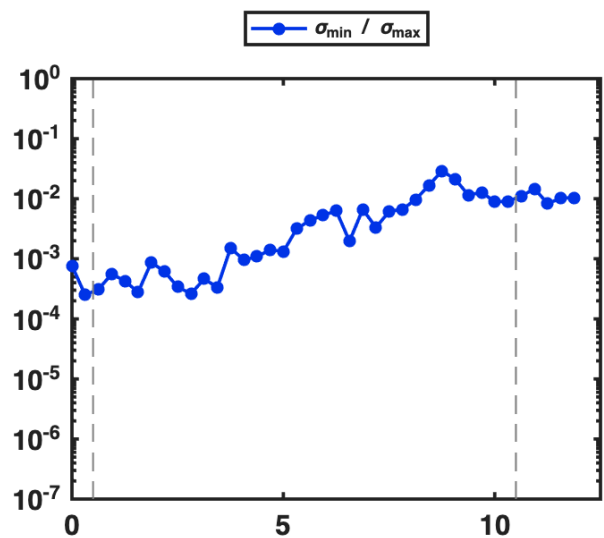
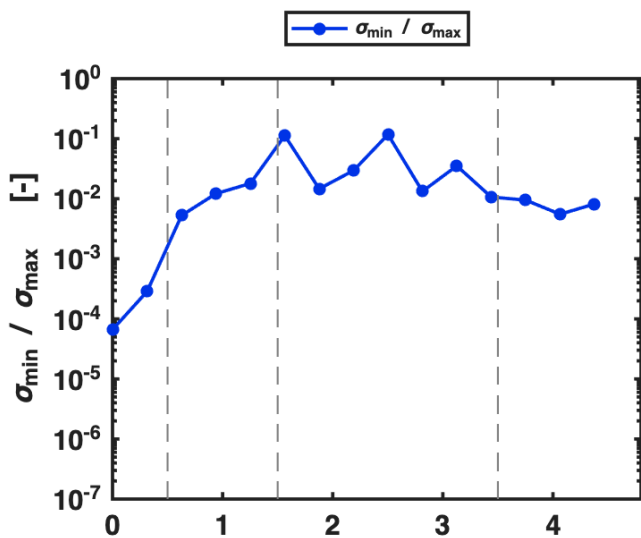
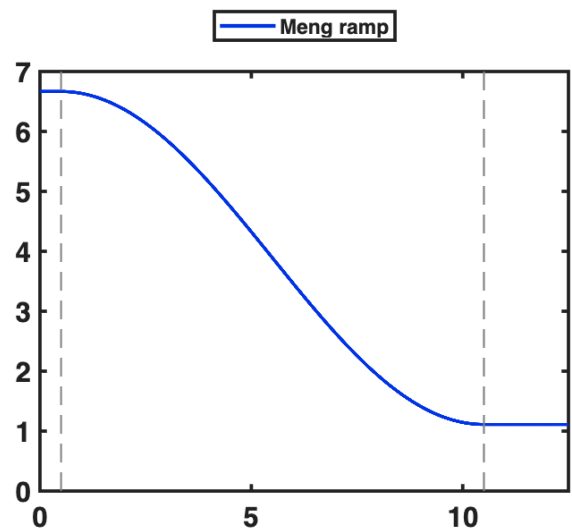
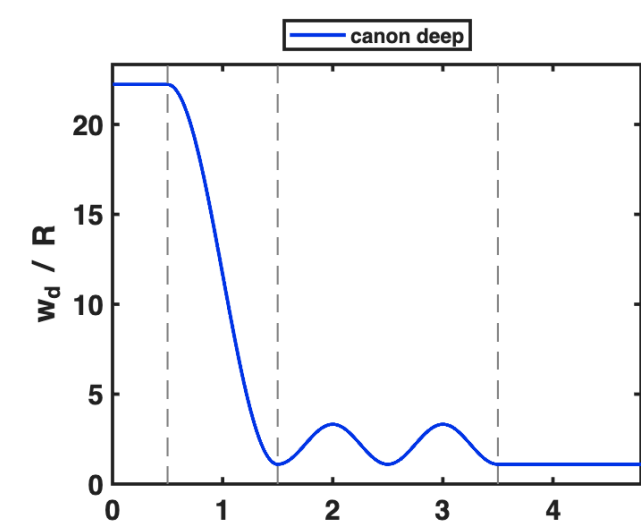
$$F_e[k]_{:,j} = I_{:,j}, \quad H[k]_{:,j} = 0 \quad \forall k \Rightarrow O_w(k_0, N)_{:,j} = 0 \Rightarrow \sigma_5 = 0$$

$$\hat{\mathbf{e}}[k_0] = \sum_{i=1}^5 \frac{\mathbf{u}_i^\top \mathbf{Y}_w}{\sigma_i} \mathbf{v}_i$$

$$\mathbf{v}_i^\top (\hat{\mathbf{e}}[k_0] - \mathbf{e}[k_0]) = \frac{\mathbf{u}_i^\top \mathbf{N}_w}{\sigma_i}, \quad \mathbb{E} \left[\left(\mathbf{v}_i^\top (\hat{\mathbf{e}}[k_0] - \mathbf{e}[k_0]) \right)^2 \right] = \frac{1}{\sigma_i^2}$$

$$\mathbf{v}_5 \rightarrow [0, 0, 0, 0, 1]^\top \Rightarrow \mathbb{E}[(\hat{e}_b - e_b)^2] \rightarrow \frac{1}{\sigma_5^2}$$

$$\frac{1}{\sigma_5} \leq \sqrt{P_{55}[0]}, \quad \sqrt{P_{55}[0]} = \sup_{\bar{w} \in \text{env}} \left| b_{\text{true}}(\bar{w}) - \frac{8}{9} \right|$$



time [s] window start k_0 , N = 500 steps

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